

Single Player Yahtzee Simulation

Abstract

The aim of this project was to develop and simulate multiple strategies for a classic dice game called 'Yahztee'. Using Python, three different strategies were implemented: 'Random Strategy' which used a simple random variate generator in Python to make all the decisions in the game, 'Spamming Yahtzee Strategy' which tried to achieve 'Yahtzee' (5 die having the same number) each turn, and 'Greedy Strategy' which tried scoring the maximum score each turn with no future turn strategy. The simulation was run for 100,000 iterations to get a better idea of the expected strategy performance instead of the results being biased due to extreme cases. The results showed that the 'Greedy Strategy' had the highest average total score of 157.86 and narrowly outperformed the 'Spamming Yahtzee Strategy' which had a very close average total score of 156.14. The 'Random Strategy' performed significantly worse, relative to the other strategies, with an average total score of 45.98. These simulations gave a slight insight into the importance of rule-based decision making and how it could affect the total score in a dice related game of chance such as Yahtzee.

Background

Yahtzee is a common dice game that was first publicized in 1956. ^[1] It can be played in both single player mode and multiplayer mode but for this project only the single player mode was looked at to focus on the simulation of playing strategy. The game involves rolling five dice with the objective of putting a score into one of thirteen scoring categories corresponding to different dice combinations. The player can reroll any number of dice up to two times per turn and the final combination would be put in one of the scoring categories shown in Figure 1 as long as the category was not already filled from a previous turn.

Upper Section	What to Score	Lower Section	What to Score
Aces (Ones)	Total of Aces only	3 of a Kind	Total of all 5 dice
Twos	Total of Twos only	4 of a Kind	Total of all 5 dice
Threes	Total of Threes only	Full House	25 points
Fours	Total of Fours only	Small Straight	30 points
Fives	Total of Fives only	Large Straight	40 points
Sixes	Total of Sixes only	YAHTZEE (5 of a Kind)	50 points
		Chance	Total of all 5 dice

Figure 1: Scoring Rules of Yahtzee ^[2]

One thing to note is that if the scoring criteria is not met and the scoring category is picked anyway, that category would then have a score of 0. The aim of the game is to have the highest total score at the end of the game when all thirteen scoring categories have been filled. While the game does involve a large element of luck in terms of what dice combinations a player rolls, there are different strategies such as deciding when and which dice to reroll and which scoring category to assign a score to. These strategies can have an impact on a player's

performance in the game. This project involved looking at three simple strategies, which were implemented in Python. To obtain a reliable estimate of the average performance of each strategy and reduce the influence of extreme cases each strategy was simulated for 100,000 games. This report will have a brief overview of the logic of each of the strategies implemented followed by a performance comparison across the strategies to draw conclusions and suggest any potential improvements.

Methodology & Results (Findings)

The three strategies that were implemented were 'Random Strategy', 'Spamming Yahtzee Strategy' and 'Greedy Strategy'. The description of each strategy and the logic used to code them are as follows:

- Random Strategy – This strategy aimed to make random decisions for every decision in the game, namely: whether to reroll the dice or not, which dice to reroll and which category to score in. These random decisions were simulated using `randint()` function from the random library which gives a random number in a certain range which would then have a corresponding decision. There is no logic in the decision making of this method and its purpose is to serve as a baseline to see how a player would do if playing without any logic. It would also allow us to see whether using simple strategies that are rule based would improve a player's overall score.
- Spamming Yahtzee Strategy – This strategy aimed to try and get the 'yahtzee' scoring category every turn. This would be achieved by having each of the five dice have the same number. This was a strategy I would use when I was new to the game and was trying to get the 'maximum' every turn and get the Yahtzee bonus (getting Yahtzee again after successfully scoring in the Yahtzee category in a previous turn) as many times as possible. The main driving force behind this strategy was the logic for rerolls of dice. If there was any majority in the numbers rolled those die would be kept and every other die would be rerolled in an attempt to get 'yahtzee'. For the scoring part if 'yahtzee' was not possible or was already scored in, the algorithm would pick the scoring category that gave the maximum points that turn. If all categories would score 0 that turn the algorithm would randomly pick any category except for 'yahtzee' in an attempt to keep it an available option for any potential future 'yahtzees'.
- Greedy Strategy – This strategy aimed to score in the maximum category each turn. This algorithm would try and chase the maximum score each turn without worrying about any future strategy. The reroll logic was to try and get a combination that would get the highest points based on the first roll combination. The logic was as follows: if the first roll had 3 or more of the same number those dice would be kept and the rest rerolled to try and get 'yahtzee' which would score the maximum of 50 points a turn (excluding bonuses.) Otherwise, the algorithm would check if there was a possibility of a large or small straight which would score 40 or 30 points respectively. For this the algorithm would keep all the dice which had successive numbers (with a minimum of 3 successive numbers) and reroll the remaining ones. For scoring the algorithm would always pick the category that had the highest score in that turn and if all categories would score 0 it would randomly pick any category from the ones still available.

Once all the strategies were completed, the simulation was run for 100,000 iterations for each of them and the resulting scorecards were recorded and analysed. The average total scores of each strategy were plotted in a bar graph to get an overview of which strategy was better. This bar graph can be seen in Figure 2 below.

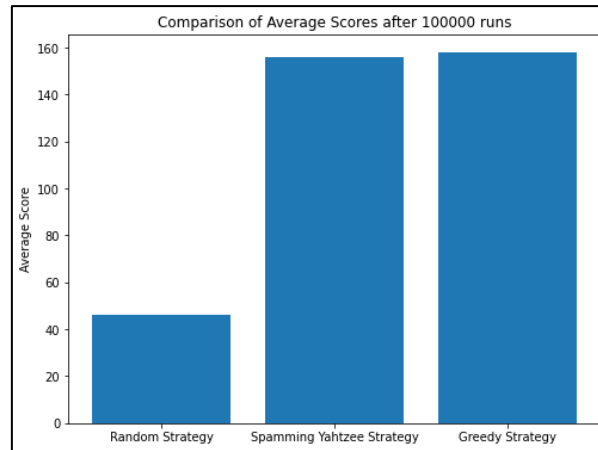


Figure 2: Comparison of Average Total Scores

We can see from the graph that there was large difference in the number of average total points scored by the random strategy algorithm and the other two. The ‘Spamming Yahtzee Strategy’ and ‘Greedy Strategy’ had very similar average total scores but the ‘Greedy Strategy’ narrowly performed better.

To understand the performances better, a few performance metrics were recorded for each of the strategies and presented in Table 1 below.

Table 1: Performance Metrics of All 3 Strategies

	Overall Results Using Different Strategies (100,000 Runs)		
	Random Strategy	Spamming Yahztee Strategy	Greedy Strategy
Average Total Score	45.98	156.14	157.86
Standard Deviation	18.25	62.38	53.55
Highest Score	160	734	616
Lowest Score	6	48	43
Number of Upper Bonuses	0	2921	797
Number of Yahtzees	63	45791	34767
Number of Yahtzee Bonuses	0	14045	8622

The ‘Random Strategy’ consistently underperformed with an average total score of 45.98. Even the highest score of the ‘Random Strategy’ was around the average total score of the other two strategies. It had the lowest standard deviation amongst the strategies, and this was likely due to many categories scoring low values or even zeroes as there was no logic behind the decision making. An interesting thing was that this strategy still managed to score 63 yahtzees (0.063%) which showed that even randomness alone can produce wanted outcomes completely by chance. However, the overall performance showed that having a

strategy even a very simply one would likely be better than complete randomness and the game 'Yahtzee' is not purely based on luck even if it does play a significant role.

The 'Spamming Yahtzee Strategy' had an average total score of 156.14 with a standard deviation of 62.38 which was the highest amongst the three strategies. This strategy got the most bonuses as well and a 'Yahtzee' percentage of 45.79% suggesting it did well. The highest score of this strategy was 734 which was the best score across the implemented strategies. Since, this method had the highest variability we can see that this strategy can be thought of as a high-risk high reward strategy.

The 'Greedy Strategy' had the highest average total score of 157.86 with a standard deviation of 53.55 meaning it had less variability than the 'Spamming Yahtzee Strategy' and can therefore be thought of being more consistent. It had a 'Yahtzee' percentage of 34.77% and overall had less bonuses compared to the 'Spamming Yahtzee Strategy'. From these metrics we can say that the 'Greedy Strategy' is a more consistent performer however has less chance of having the highest possible score over a large number of trials compared to the 'Spamming Yahtzee Strategy'.

For further analysis, the average score of each of the scoring categories was recorded and is shown in Table 2 below.

Table 2: Average Score for Each Scoring Category Across all Runs

Scoring Category	Average Score Using Different Strategies (100,000 Runs)		
	Random Strategy	Spamming Yahztee Strategy	Greedy Strategy
<i>Aces</i>	0.832	1.767	1.536
<i>Twos</i>	1.655	4.188	3.735
<i>Threes</i>	2.504	6.267	5.699
<i>Fours</i>	3.352	8.151	7.304
<i>Fives</i>	4.156	9.867	8.946
<i>Sixes</i>	5.021	11.590	10.593
<i>Upper Section Bonus</i>	0.000	1.022	0.279
<i>Three of a Kind</i>	3.711	17.638	17.521
<i>Four of a Kind</i>	0.351	16.264	14.340
<i>Full House</i>	0.944	16.936	11.583
<i>Small Straight</i>	4.612	7.667	28.981
<i>Large Straight</i>	1.233	0.131	3.397
<i>Yahtzee</i>	0.032	22.896	17.384
<i>Chance</i>	17.492	17.594	17.495
<i>Yahtzee Bonus</i>	0.000	14.045	8.622

As expected, the 'Random Strategy' scores the lowest in all categories compared to the other two strategies. Something to highlight is that the chance category has a similar score across all the strategies. A likely reason for this is that there is no required pattern or rule for scoring in the chance category and it is always the sum of the five dice in the final combination and so all strategies are almost equally likely to score similarly in that category.

Interestingly, compared to the 'Greedy Strategy', the 'Spamming Yahtzee Strategy' had a better average score across all scoring categories except small straight and large straight. This is due to the rerolling logic of both strategies. Since, the 'Spamming Yahtzee Strategy' consistently aims to roll five of the same number, it automatically boosts the likelihood of having a higher score in all upper categories as well as three of a kind, four of a kind, full house and Yahtzee due to the scoring mechanics of those categories since they involve repeating numbers. However, this logic would not work for the small or large straight categories as they require sequential numbers which is targeted by the 'Greedy Strategy's' reroll algorithm.

Histograms of the three strategies were then plotted and are shown in Figure 3 below.

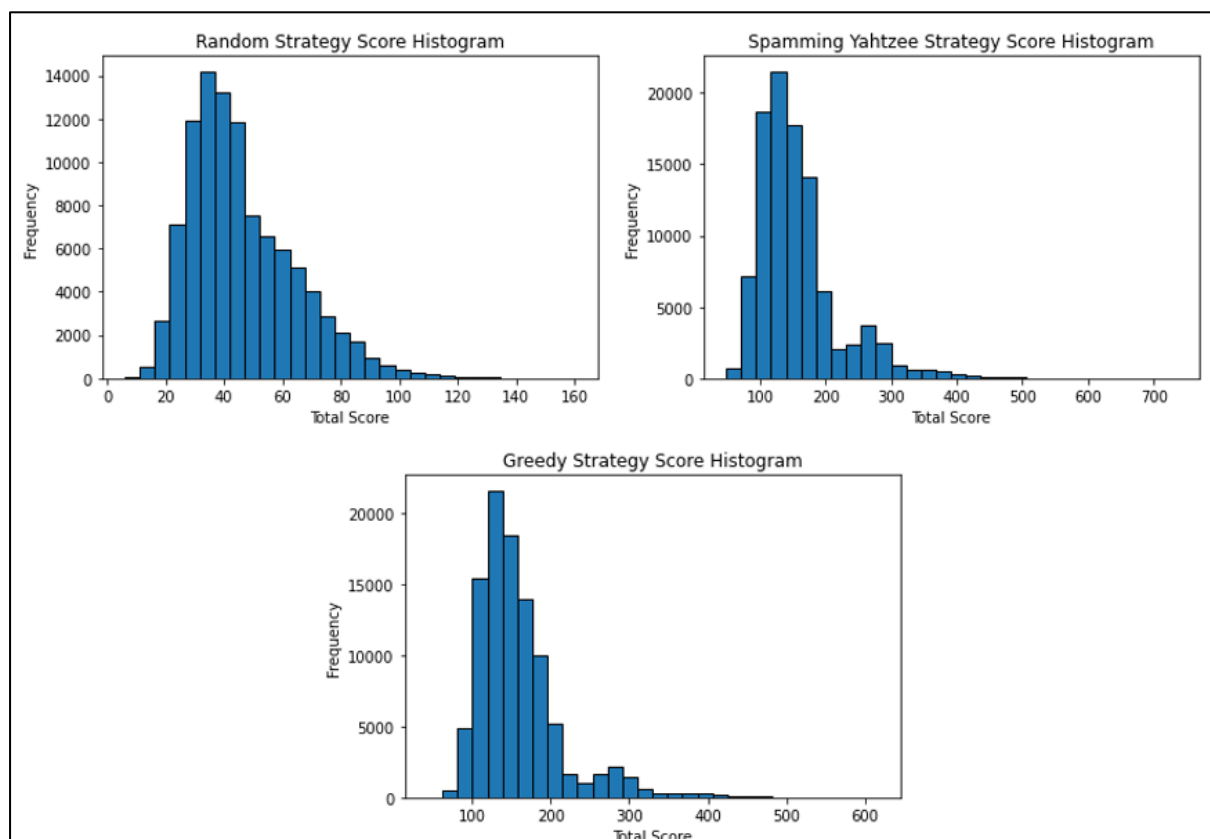


Figure 3: Histogram of Scores of the Three Strategies

We can see from the histograms that all three distributions are right-skewed. The 'Random Strategy' has the least spread however has the lowest peak. The histograms of the 'Spamming Yahtzee Strategy' and 'Greedy Strategy' show similar distributions with the 'Spamming Yahtzee Strategy' having a higher spread. We can see that there is a slight increase in the frequency around the 270-300 score which is possibly due to the bonuses such as upper bonus or yahtzee bonus. Overall, the histograms reinforce the earlier analysis that the 'Random Strategy' is the least effective of the three, the 'Spamming Yahtzee Strategy' has higher variability but may score higher occasionally and the 'Greedy Strategy' is a relatively consistent and better performer on average.

To compare how the performance metrics change for the 'Spamming Yahtzee Strategy' and the 'Greedy Strategy' based on the number of runs the comparison was recorded in Table 3.

Table 3: Comparison of Metrics using Different Number of Runs

	Comparison of Results Using Different Number of Runs					
	1000 Runs		10,000 Runs		100,000 Runs	
	Spamming Yahtzee Strategy	Greedy Strategy	Spamming Yahtzee Strategy	Greedy Strategy	Spamming Yahtzee Strategy	Greedy Strategy
Average Total Score	153.60	158.38	154.73	156.94	156.14	157.86
Standard Deviation	58.76	53.28	61.03	51.95	62.38	53.55
Upper Bonuses %	1.8%	0.5%	2.67%	0.89%	2.92%	0.79%
Yahtzee %	47%	34.2%	45.8%	34.19%	45.79%	34.77%
Yahtzee Bonuses %	13.1%	9.3%	13.69%	8.75%	14.05%	8.62%

We can see from the table above that the ‘Spamming Yahtzee Strategy’ had a slight increase in average total score when increasing the runs from 1000 to 100,000 runs while also having an increase in standard deviation. On the other hand, the ‘Greedy Strategy’ had a more stable average score and standard deviation as the runs increased. Once again, the high variability for the ‘Spamming Yahtzee Strategy’ can be seen from this data.

Conclusion

In conclusion, from the three strategies that were tested the ‘best’ strategy in terms of consistency and overall highest average total score was the ‘Greedy Strategy’. The ‘Spamming Yahtzee Strategy’ had the highest score in all the runs across strategies but also had the highest variability which indicates it is a high-risk high reward strategy. The ‘Random Strategy’ had the worst performance and showed that even a simple strategy will on average be much better than playing completely randomly.

One thing that could be affecting the performance of the ‘Greedy Strategy’ could be that the algorithm randomly assigns a scoring category if the score across all available categories is zero. This was done because this strategy was only trying to maximize the current turn without considering any future consequences. A better strategy could be a combination of the ‘Spamming Yahtzee Strategy’ and the ‘Greedy Strategy’ by using the scoring logic of the former and the reroll logic of the latter. Keeping high scoring categories open when a zero needs to be scored would allow the chance of scoring higher in upcoming turns. This hybrid approach could balance the consistency of the ‘Greedy Strategy’ with the opportunistic scoring of the ‘Spamming Yahtzee Strategy’. Additionally, incorporating probability into decision making such as estimating the likelihood of getting specific combinations in upcoming turns may further improve performance. This would enable the algorithm to think of future consequences when choosing which category to ‘sacrifice’ when a zero needs to be added.

References

1. <https://en.wikipedia.org/wiki/Yahtzee>

2. <https://www.hasbro.com/common/instruct/yahtzee.pdf>
3. https://www.youtube.com/watch?v=x-Ag2_bJ40Y